

2024



Harnessing AI and ML for Identity Verification Using Amazon SageMaker and Amazon Rekognition with Jumio

Learn how Jumio is using Amazon SageMaker and Amazon Rekognition to deliver a seamless and secure identity verification experience.

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Overview

When people log on to online bank accounts, healthcare portals, and gambling sites, they need a reliable way to identify themselves. Using Amazon Web Services (AWS), identity verification (IDV) company [Jumio](#), an [AWS Partner](#), provides several frictionless IDV solutions to meet the unique needs of customers around the world. And because many criminals are now using advanced tools to commit fraud, Jumio is harnessing the power of artificial intelligence (AI) and machine learning (ML) to stay ahead of the threat. “Our goal is to deliver solutions that convert as many legitimate customers as possible while identifying as much fraud as possible,” says Daryl Huff, vice president of advanced development at Jumio.

Today, many transactions are completed online, and organizations must put guardrails in place to remain compliant with relevant regulations and safeguard against fraud. For example, online casinos are required to verify that people who want to gamble are of legal age. Similarly, people who want to rent a scooter for an hour or an apartment for a weekend often use IDV to make their transactions more secure. Jumio strives to support a wide variety of use cases for its customers. “In total, we work with over 5,000 ID types in over 200 countries and territories,” says Loryll DeNamur, director of corporate marketing at Jumio.



Opportunity | Jumio Uses AWS to Seamlessly Verify People

Founded in 2010 and based in Sunnyvale, California, Jumio has pioneered techniques that are now a mainstay across the IDV space. For example, the company offers a seamless end user experience that makes it simple for people to verify their identities by photographing their IDs and taking a selfie. Businesses around the world now rely on this method to verify that people are who they say they are.

According to the NSA and the FBI, the generative AI market is expected to hit \$100 billion by 2030, putting advanced capabilities into the hands of cybercriminals. This prediction makes state-of-the-art IDV procedures critically important for Jumio and its customers. “We have over 50 ML models involved in the IDV process, determining whether the people on-screen are live humans or some type of deepfake or screen attack,” says Huff.

Jumio built its solution on AWS, and it offers customers a range of solutions to meet their specific use cases. For example, Jumio offers both ID verification and IDV through the [Jumio KYX Platform](#). However, if customers need only ID verification, they don’t have to pay for IDV. This flexibility makes it simple for customers to optimize the tradeoff between security and the end user experience. Banks, for example, might choose stringent security procedures, knowing that their end users value the added protection. A university, however, might assess that top-tier security measures don’t justify the added friction for their end users.

Solution | Advancing IDV Innovation Using Amazon Rekognition and Amazon SageMaker

Jumio innovates to stay one step ahead of cybercriminals, using two services from AWS to help power its offerings.

Building 50-Plus ML Models at Scale Using Amazon SageMaker

To validate the authenticity of government-issued IDs, Jumio uses [Amazon SageMaker](#), a service to build, train, and deploy ML models for any use case with fully managed infrastructure, tools, and workflows. “We build and train models of all sizes and purposes using Amazon SageMaker,” says Huff. “Given the variety of systems that AWS offers, we tune our development training environment to match the needs of each model in a cost-efficient way.”

The company’s engineers are constantly experimenting and enhancing its solutions using Amazon SageMaker. “We’re starting to use some of the large language models for new purposes,” says Huff. “We have proprietary models and analysis techniques that look for inconsistencies in the extracted data



tampering or forgery of the identity document. We're also extracting text from IDs and comparing it against government databases."

Providing Secure Face Verification Using Amazon Rekognition

Many Jumio customers require biometric verification as part of their IDV process. As part of its overall solution, Jumio also deploys [Amazon Rekognition](#), which offers pretrained and customizable computer vision capabilities to extract information and insights from images and videos. By applying computer vision to the verification process, Jumio strengthens security while delivering a virtually seamless user experience.

Using Amazon Rekognition, Jumio also advances its mission of providing fair IDV. "One of our core value propositions is to offer solutions that are free from bias and work across all different genders, age ranges, and ethnicities," says Huff. "We find Amazon Rekognition to be highly accurate, and it's key to advancing our mission."

When it comes to face verification, subtle details matter. Amazon Rekognition helps Jumio catch those details. "You can hold a passport for 10 years, and at the end of that period, you're 9 years older than you were in your passport photo. A critical part of our business is to make sure that the older person in front of the camera matches the image from the government-issued document. And that's where Amazon Rekognition 1:1 and 1:n matching helps," says Huff.

Outcome | Streamlining IDV around the World on AWS

More broadly, Jumio has committed to going all in on AWS to increase the impact of its offerings. Because [AWS Regions](#) are available in many global geographies, Jumio advances its data sovereignty compliance and scales around the world by using AWS.

The company also scales its services by offering a streamlined onboarding experience. "Our solution converts as many of the good customers while identifying as many of the bad customers as possible," says Huff. "Providing our customers with the ability to find that balance is one of our core values."

About Jumio

From account opening to ongoing monitoring, Jumio's platform provides AI-driven identity verification, risk signals, and compliance solutions that help organizations accurately establish, maintain, and reassert trust.

AWS Services Used

Amazon Rekognition

Amazon Rekognition offers pre-trained and customizable computer vision (CV) capabilities to extract information and insights from your images and videos.

[Learn more »](#)



Amazon SageMaker

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